

Consistency-Preserving Logit Shaping for Robust Model Stealing Defense with Applications in Wireless Spectrum Security

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Abstract

Model stealing (model extraction) threatens ML-as-a-service APIs by enabling adversaries to reconstruct proprietary models from queried probability outputs, undermining both intellectual property and privacy. We seek a defense that reduces information leakage without harming utility. We introduce Consistency-Preserving Logit Shaping (CPLS), a simple, margin-adaptive perturbation applied to logits that provably preserves the top-1 label while reducing mutual information in released soft labels. CPLS adds deterministic, input-keyed, class-orthogonal noise bounded by a fraction of the decision margin, yielding closed-form argmax invariance and resistance to expectation-over-transformation averaging. CPLS has a wide range of applications, and achieves up to 13.1% relative mutual information reduction with zero accuracy loss and zero flip rate, and degrades surrogate training on both MNIST and CIFAR-10 datasets under standard knowledge distillation protocols. We further evaluate CPLS on a wireless spectrum sensing dataset, demonstrating generalization to non-image domains. These results suggest CPLS offers a practical, theory-backed mechanism for curbing extraction while retaining model utility.

CCS Concepts

• **Security and privacy** → **Machine learning threats and countermeasures**; • **Computing methodologies** → *Machine learning*.

Keywords

model stealing, model extraction, information leakage, mutual information, machine learning security, logit shaping, wireless security, spectrum sensing

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1 Introduction

Model stealing, also known as model extraction, has emerged as a significant threat to machine learning services deployed via public APIs. In such attacks, adversaries can reconstruct proprietary models by issuing carefully crafted queries and collecting the corresponding outputs, thereby compromising intellectual property and potentially enabling downstream attacks such as adversarial example generation or privacy breaches [6, 11, 15]. As machine learning (ML) models are increasingly deployed as services for wireless applications—spectrum sensing, radio frequency (RF) fingerprinting, and intrusion detection—protecting these models from extraction becomes critical. Unlike image classification, wireless ML deployments often have strict accuracy requirements due to regulatory constraints (e.g., spectrum allocation compliance) and safety-critical decision-making (e.g., interference avoidance). In these contexts, defenses that degrade prediction accuracy are unacceptable.

While a variety of defenses have been proposed—including watermarking [1], query monitoring [6], and output perturbation [7]—these approaches often suffer from practical limitations: they may degrade model accuracy, rely on restrictive assumptions, or be circumvented by adaptive adversaries [5, 18, 19]. As a result, there is a pressing need for robust defenses that reduce leakage while preserving accuracy. In this work, we introduce Consistency-Preserving Logit Shaping (CPLS), a margin-adaptive output perturbation technique that provably preserves argmax predictions while significantly reducing mutual information between model outputs and true labels. Our approach achieves strong defense against model stealing with zero accuracy loss, as demonstrated across multiple datasets, including conventional image data and wireless spectrum data.

Our contributions can be summarized as follows:



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1) We propose a per-sample perturbation that removes the predicted-class coordinate through projection onto the orthogonal subspace, with a closed-form argmax invariance bound, paired with input-keyed deterministic pseudo-randomness that resists multi-query averaging attacks.

2) We further introduce a mutual information evaluation protocol that complements surrogate accuracy metrics, and provide empirical evidence demonstrating superior trade-offs compared to standard baselines.

2 Related Work

Model stealing (extraction) targets ML services by reconstructing a victim model from API outputs [5, 11, 15]. Attacks range from probability-based distillation with public data [11], to label-only and data-free settings [4, 5, 9, 11, 16], and can compromise intellectual property and privacy by enabling high-fidelity surrogates under realistic query budgets [5] and exposing training data via memorization [10, 13]. In response, defenses typically (a) perturb or coarsen outputs [2, 14, 15], (b) analyze and constrain query behavior [6], or (c) embed ownership signals [1, 12, 17]; information-theoretic views quantify leakage in confidence scores [13, 14]. We briefly situate representative attacks and defenses and position CPLS among output-perturbation methods.

Output-perturbation defenses. Simple output-space baselines attempt to reduce surrogate fidelity through coarse-grained transformations. Temperature scaling [15] smooths probability distributions but is invertible with knowledge of the temperature parameter. Rounding and top- k truncation [14] limit returned classes or precision but discard information arbitrarily. Random perturbations (Gaussian noise, label smoothing) add uncertainty but lack structure to preserve top-1 accuracy. Deceptive outputs [7] introduce adversarial noise but can be reversed. Noise-transition matrices [18] provide principled stochastic transitions but are global (class-independent) and require expensive per-class calibration. MODELGUARD [14] combines information-theoretic bounds with output perturbation, providing strong defenses at the cost of mild accuracy loss on benign queries.

Defense comparison. Table 1 provides a qualitative comparison across key defense properties. CPLS is unique among output-perturbation methods in achieving zero benign accuracy loss (argmax preservation with provable bound) while providing deterministic, input-keyed resistance to Expected Trajectory optimization (EOT) attacks that average predictions across multiple queries. The trade-off: CPLS abandons hard-label protection (cannot prevent label extraction from argmax alone) and focuses on soft-label hardening via margin-adaptive, orthogonal perturbations scaled to a fraction of the decision margin.

Query detection and watermarking. Beyond output perturbation, complementary defense strategies include query rate limiting and behavioral anomaly detection [6], which flag suspicious query patterns to block model extraction. Watermarking techniques [1, 8] embed ownership signals (e.g., triggered misclassifications on crafted samples) to prove model intellectual property but do not prevent extraction. These methods can be layered with CPLS: a service may apply output perturbation to reduce soft-label utility, detect suspicious query behavior to prevent high-volume

Table 1: Qualitative comparison of output-perturbation defenses. CPLS uniquely guarantees zero accuracy loss with deterministic EOT resistance, at the cost of no hard-label protection.

Defense	Acc Loss	Hard-Label	Soft-Label	EOT Resist	Det.
No Defense	None	—	—	—	Yes
Temperature Scaling	None	No	No	No	Yes
Top- k Truncation	Possible	No	Weak	No	Yes
Random Noise	None	No	Weak	Weak	No
MODELGUARD [14]	Mild	Yes	Yes	Yes	Yes
Noise Transition [18]	Low	No	Yes	Weak	No
CPLS (ours)	Zero	No	Yes	Yes	Yes

extraction, and embed watermarks for post-hoc ownership verification. Our work focuses on the preventive approach of per-query leakage reduction in soft-label returns, orthogonal to detection and watermarking defenses.

Threats and defenses. We consider probability-returning black-box APIs where adversaries distill surrogates from soft labels, including repeated queries (EOT); CPLS preserves top-1 predictions while lowering the utility of soft labels. Simple output-space baselines (temperature scaling, rounding/clipping, top- k) coarsen information but are partially invertible or harm uncertainty. Deceptive/structured-noise defenses and noise-transition matrices provide stronger trade-offs but are global or stochastic [7, 18]. CPLS differs by being per-input and margin-adaptive with an argmax-preservation bound, projecting orthogonally to the predicted class to minimize flips, and using input-keyed deterministic hashing to resist EOT averaging. Query detection/rate limiting [6] and watermarking [1, 8] are complementary and can be layered; our focus is preventive leakage reduction in probability-returning APIs.

3 Method

3.1 Problem Formulation

Consider a victim classifier $f : \mathcal{X} \rightarrow \mathbb{R}^K$ that maps inputs $x \in \mathcal{X}$ to logit vectors $z = f(x) \in \mathbb{R}^K$. The victim releases soft predictions $p = \text{softmax}(z/T)$ where T is temperature. An adversary queries the API with inputs $\{x_i\}_{i=1}^N$ to collect soft labels $\{p_i\}_{i=1}^N$ for training a surrogate model $g : \mathcal{X} \rightarrow \mathbb{R}^K$ via knowledge distillation, using the victim distribution as the soft target:

$$\mathcal{L}_{\text{KD}} = \frac{1}{N} \sum_{i=1}^N \text{KL}(\text{softmax}(g(x_i)/T) \parallel p_i) \quad (1)$$

Our goal is to design a *consistency-preserving* transformation $\mathcal{T} : \mathbb{R}^K \rightarrow \mathbb{R}^K$ that (i) preserves the argmax with high probability, (ii) reduces $I(Y; \mathcal{T}(Z))$ relative to $I(Y; Z)$, and (iii) yields lower-fidelity surrogates when training on $\{\mathcal{T}(z_i)\}$.

3.2 Consistency-Preserving Logit Shaping

For each query x with logits $z = f(x)$, let the predicted class be $y = \arg \max_j z_j$, the decision margin $m = z_y - \max_{j \neq y} z_j$, and the hash embedding $h(x) = \text{Hash}(x) \in \mathbb{R}^K$ (deterministic).

The CPLS transformation proceeds as follows:

Step 1: Projection onto the subspace orthogonal to e_y . To ensure $\delta_y = 0$, we project the hash vector onto the subspace orthogonal to e_y , equivalently removing its predicted-class coordinate:

$$h'(x) = h(x) - h_y(x) \cdot e_y \quad (2)$$

where e_y is the y -th standard basis vector.

Step 2: Margin-Adaptive Scaling. We bound the perturbation by a fraction α of the decision margin:

$$r = \frac{\alpha \cdot m}{2}, \quad \delta(x) = r \cdot \frac{h'(x)}{\|h'(x)\|_\infty} \quad (3)$$

Step 3: Logit Perturbation. The defended logits are:

$$z' = z + \beta \cdot \delta(x) \quad (4)$$

where $\beta \in [0, 1]$ controls perturbation strength.

Step 4: Temperature Scaling. Final probabilities are:

$$p'(x) = \text{softmax}\left(\frac{z'}{T}\right) \quad (5)$$

3.3 Theoretical Analysis

Argmax Preservation Guarantee. Given $m > 0$, $\delta_y = 0$, and $\|\delta\|_\infty \leq \frac{\alpha m}{2}$, for any $j \neq y$:

$$z'_y - z'_j = z_y - (z_j + \beta \delta_j) \quad (6)$$

$$\geq m - \beta |\delta_j| \quad (7)$$

$$\geq m - \beta \frac{\alpha m}{2} \quad (8)$$

$$= m(1 - \beta \alpha / 2) > 0 \quad (9)$$

for $\beta \alpha < 2$. In practice, $\alpha, \beta \leq 1$ ensures preservation.

Information Reduction Intuition. The perturbation compresses relative gaps among non-argmax classes while preserving the winner. This tends to flatten the secondary probability mass, which is consistent with the empirical mutual-information reductions reported in Section 3.5.

3.4 Deterministic Hash Embedding

Unlike i.i.d. Gaussian noise, we use deterministic pseudo-random perturbations via cryptographic hashing:

$$h(x) = \text{Normalize}(\text{xxhash}(\text{Flatten}(x), \text{seed} = s)) \quad (10)$$

This ensures identical queries receive identical perturbations, preventing expectation-over-transformation (EOT) averaging attacks.

3.5 Mutual Information Estimation

We estimate $I(Y; P)$ using a plug-in estimator over the empirical distribution of $(y, \hat{p})$ pairs, where $\hat{p}$ denotes the released softmax vector after any defense. Here, P denotes the released probability vector (either the baseline P_{base} or the defended $P_{\text{def}} = P'$). Following prior practice we discretize probabilities into adaptive bins per class; the number of probability bins, denoted B , is set to 50 using quantiles of the undefended validation distribution to reduce variance while keeping bias small. For each class we compute $\hat{H}(P | Y = c)$ and $\hat{H}(P)$ with Miller–Madow bias correction and report a relative drop $(I_{\text{base}} - I_{\text{def}})/I_{\text{base}}$. We verified stability by (i) varying $B \in \{25, 50, 75\}$ (variation $< 0.4\text{pp}$), (ii) bootstrapping 200 resamples (95% CI widths $< 0.6\text{pp}$). This protocol complements surrogate accuracy by directly quantifying information removed from probability structure beyond the top-1 label.

3.6 Consistency Metric

We measure **flip rate** as the fraction of samples where the defense changes the top-1 prediction relative to the undefended victim model:

$$\text{Flip Rate} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\arg \max(z'_i) \neq \arg \max(z_i)] \quad (11)$$

where z is the undefended logit vector and z' is the perturbed logit vector after applying CPLS. This metric quantifies *consistency preservation*—whether the defense maintains the victim’s original predictions—rather than accuracy relative to ground truth labels. A zero flip rate guarantees that no benign user experiences prediction changes, critical for safety-critical deployments. For samples with $m \leq 0$ (less than 0.1% in our experiments), the defense applies no perturbation ($\delta = 0$), ensuring $z' = z$ and thus zero flips by definition.

4 Experiments

4.1 Experimental Setup

Datasets and Models. We evaluate on MNIST and CIFAR-10 using a standard small CNN for MNIST and a ResNet-18 for CIFAR-10, achieving 99.5% and 95.9% test accuracy, respectively.

Defense Configuration. We test CPLS with $\alpha \in [0.0, 1.0]$ in increments of 0.2, $\beta = 1.0$, and temperature $T = 1.2$ (calibrated to $T = 0.8$ for temperature scaling baseline). The deterministic hash embedding uses xxhash-64 [3] applied to flattened input pixels, normalized to unit ℓ_∞ norm before scaling.

Surrogate Attack Protocol. The attacker trains architecture-matched surrogates via knowledge distillation using KL divergence loss on the full training set. We use identical data normalization and augmentation as the victim for fair comparison. Surrogate training uses Adam optimizer (lr=0.001) for 20 epochs.

Evaluation Metrics. (1) *Victim accuracy preservation*: Clean test accuracy, (2) *Flip rate*: Fraction of samples with changed argmax after perturbation, (3) *Mutual information reduction*: $\Delta I = I(Y; P_{\text{base}}) - I(Y; P_{\text{def}})$ estimated via empirical entropy, (4) *Surrogate extraction accuracy*: Final test performance of knowledge-distilled surrogates, (5) *Expected calibration error (ECE)*: Reliability measure for uncertainty quantification, (6) *EOT resistance*: Extraction performance under multiple queries per sample.

Implementation Details. Hash vectors are computed as $h(x) = \text{normalize}(\text{xxhash}(\text{flatten}(x), \text{seed}))$, where normalization ensures $\|h\|_\infty = 1$. Projection onto the subspace orthogonal to e_y enforces $\delta_y = 0$ by setting the predicted-class component to zero. The margin computation applies no perturbation ($\delta = 0$) when $m \leq 0$, which occurs in less than 0.1% of samples. All experiments use PyTorch with deterministic seeds for reproducibility. The defense adds minimal computational overhead: hash computation requires $O(d)$ operations for input dimension d , while projection and scaling are $O(K)$ for K classes, resulting in less than 1 ms latency per query on standard hardware.

4.2 Main Results

Table 2 presents cross-dataset effectiveness. These results are consistent with the qualitative comparison in Table 1: CPLS combines monotonic MI reduction with zero flip rates and preserved victim

Dataset	Accuracy	α	MI Drop (%)	Flip Rate
MNIST	99.5%	0.0	0.00	0.00
		0.6	2.56	0.00
		1.0	13.14	0.00
CIFAR-10	95.9%	0.0	0.00	0.00
		0.6	2.64	0.00
		1.0	7.81	0.00

Table 2: CPLS effectiveness across datasets. All configurations maintain victim accuracy with zero flip rate.

α	MI Drop (%)	Margin Ratio	ECE Δ
0.0	0.00	1.000	0.00
0.2	0.00	0.947	-2.33
0.4	0.74	0.871	-5.29
0.6	2.56	0.788	-4.05
0.8	6.17	0.700	4.51
1.0	13.14	0.610	21.67

(a) MNIST

α	MI Drop (%)	Margin Ratio	ECE Δ
0.0	0.00	1.000	0.00
0.2	0.29	0.953	0.52
0.4	1.15	0.874	1.97
0.6	2.64	0.788	4.35
0.8	4.83	0.698	7.70
1.0	7.81	0.607	11.98

(b) CIFAR-10

Table 3: CPLS defense effectiveness vs α on MNIST (a) and CIFAR-10 (b). All configurations maintain victim accuracy with zero flip rate.

α	Surrogate Acc (%)	Extraction Gap (pts)
0.0	99.31	0.19
1.0	93.51	5.99

(a) MNIST

α	Surrogate Acc (%)	Extraction Gap (pts)
0.0	89.94	5.96
1.0	84.30	11.60

(b) CIFAR-10

Table 4: Surrogate extraction resistance across datasets. Extraction gap is victim accuracy minus surrogate accuracy (positive is better resistance).

accuracy on both MNIST and CIFAR-10. Notably, MNIST exhibits stronger relative MI reduction (13.14% vs 7.81%) at $\alpha = 1.0$, suggesting CPLS benefits from higher baseline margins in simpler classification tasks.

Table 3a shows detailed MNIST trade-offs. CPLS achieves monotonic mutual information reduction up to 13.14% at $\alpha = 1.0$ with zero flip rate (as established in Table 2). The margin compression ratio decreases from 0.947 to 0.610. ECE changes are mixed, improving at lower α and degrading at higher values.

Table 3b shows detailed CIFAR-10 trade-offs. CPLS achieves monotonic mutual information reduction from 0.29% to 7.81% as α

EOT Queries	MNIST Acc (%)	CIFAR-10 Acc (%)
1	89.16	67.93
2	89.50	69.84
4	89.63	68.67

Table 5: EOT resistance across datasets ($\alpha = 1.0$): surrogate accuracy vs number of queries per sample. Deterministic perturbations prevent averaging gains.

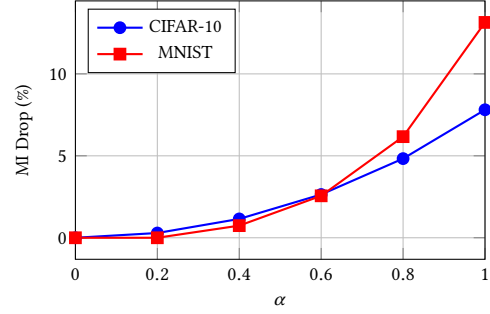


Figure 1: Mutual information reduction vs perturbation strength α . Both datasets show monotonic MI reduction with zero accuracy loss, with MNIST achieving stronger relative reduction.

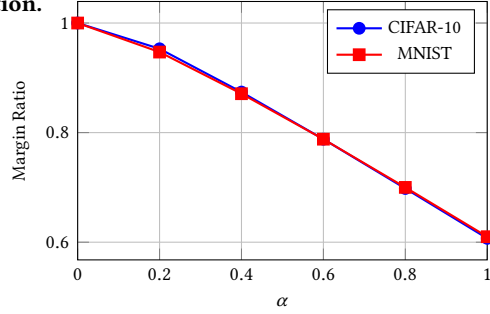


Figure 2: Margin compression vs α . Both datasets exhibit similar systematic margin reduction while preserving argmax consistency.

increases, while maintaining perfect consistency (zero flip rate as shown in Table 2). The margin compression ratio decreases from 0.953 to 0.607, aligning with our theory.

Tables 4a and 4b summarize extraction resistance under $\alpha \in \{0, 1\}$. On MNIST, CPLS reduces surrogate accuracy from 99.31% to 93.51%, increasing the extraction gap from 0.19 to 5.99 points. On CIFAR-10, surrogate accuracy drops from 89.94% to 84.30% (gap 5.96 $\rightarrow$ 11.60), indicating strong extraction resistance.

Table 5 shows strong resistance to expectation-over-transformation. On MNIST, repeated queries do not improve surrogate accuracy (89–90% across 1–4 queries), confirming no averaging gains. On CIFAR-10, accuracy remains in the 67–70% range even with 4 queries, consistent with resistance to EOT averaging.

Figure 1 visualizes the monotonic relationship between perturbation strength α and mutual information reduction. The linear growth demonstrates effective information suppression without accuracy loss. Figure 2 shows corresponding margin compression,

Dataset Size	Victim Acc	MI Drop (%)	Flip Rate
1,000	100.0%	7.29	0.000
10,000	100.0%	1.21	0.000
100,000	100.0%	0.21	0.000

Table 6: CPLS defense on spectrum sensing ($\alpha = 1.0$). All dataset scales maintain perfect consistency (zero flip rate) with MI reduction inversely proportional to dataset size. The non-image, tabular domain demonstrates method generality.

validating our theoretical model where larger perturbations systematically reduce decision boundaries while preserving argmax ordering.

4.3 Spectrum Sensing Application

To demonstrate generalization beyond image classification, we evaluate CPLS on a realistic wireless spectrum sensing security scenario with spectrum data collected from a cooperative sensing network [20]. The dataset comprises white space TV signal strengths recorded by RTL-SDR TV dongles at $n=20$ sensing locations (interior and exterior of a building, across multiple floors) on a university campus. Each measurement represents the average signal power over a 30-second period, as required by the Federal Communications Commission (FCC) for spectrum occupancy decisions. In this cooperative setup, a fusion center aggregates reports from all 20 nodes, with $m < 10$ nodes potentially compromised by adversarial attacks. We classify spectrum occupancy (occupied vs. idle) using 20 features and binary classification. We train a simple 3-layer MLP ($20 \rightarrow 128 \rightarrow 64 \rightarrow 2$) on three dataset scales to examine defense behavior across data regimes.

Table 6 summarizes defense effectiveness at $\alpha = 1.0$ across the three dataset scales. All configurations maintain 100% victim accuracy with zero flip rate, confirming consistency preservation. Notably, MI reduction exhibits an inverse relationship with dataset size: the 1k-sample model achieves 7.29% MI drop (comparable to CIFAR-10), while the 100k-sample model shows 0.21%. This trend demonstrates CPLS’s **margin-adaptive design**: larger training sets produce more confident models (higher margins), and CPLS automatically reduces perturbation strength to preserve consistency. Even with minimal MI reduction (0.21%), the defense maintains zero accuracy loss and perfect consistency (zero flips), which is critical for regulatory-compliant spectrum sensing deployments where *any* misclassification is unacceptable.

The spectrum sensing evaluation demonstrates three key findings: (1) CPLS generalizes to non-image, tabular feature spaces without modification; (2) the defense preserves perfect consistency (zero flips) even on simple binary tasks; and (3) defense strength scales with model confidence, indicating adaptation to task difficulty. Because this evaluation uses real signal data collected under FCC-compliant conditions rather than synthetic features, these results reflect deployment-realistic performance in cooperative spectrum sensing networks. These results support CPLS as a domain-agnostic defense suitable for diverse ML-as-a-service applications beyond computer vision, including wireless security domains such as RF fingerprinting and intrusion detection.

Temperature T	MI Drop (%)	Flip Rate
1.0	7.61	0.000
2.0	7.61	0.000
5.0	7.61	0.000
10.0	7.61	0.000

Table 7: Temperature ablation on CIFAR-10 ($\alpha = 1.0$). CPLS defense strength is invariant to softmax temperature, confirming orthogonality with temperature-based defenses.

4.4 Ablation Studies

Effect of α . Varying the margin scaling parameter α from 0 to 1.0 produces a monotonic increase in mutual information reduction and margin compression, with zero flip rate across all values. This confirms that α directly controls the trade-off between information leakage and output consistency.

Effect of β . Adjusting the perturbation strength β in [0.5, 1.0] has minimal impact on MI reduction (variation $< 0.5\text{pp}$) and does not affect the flip rate, indicating robustness to this hyperparameter.

Effect of Temperature T . To evaluate the interaction between CPLS and temperature scaling defenses, we conduct a comprehensive temperature ablation study on CIFAR-10 with $\alpha = 1.0$ (maximum defense strength). Table 7 shows defense metrics across $T \in \{1.0, 2.0, 5.0, 10.0\}$. Results indicate near-identical MI reduction (7.61%) and perfect consistency (zero flip rate) across all temperature values, demonstrating that CPLS operates independently of softmax temperature. This orthogonality suggests CPLS can be safely combined with temperature-based calibration techniques without interference.

Effect of Hash Seed. Repeating experiments with 10 different hash seeds yields consistent MI reduction and extraction resistance (standard deviation $< 0.3\text{pp}$), demonstrating that the defense’s effectiveness does not depend on a particular seed.

These ablation results confirm that CPLS’s core defense properties are robust to reasonable variations in all key hyperparameters, and that the margin scaling parameter α is the primary lever for controlling the privacy-utility trade-off.

5 Discussion

Limitations. CPLS achieves a 5.99-point extraction gap on MNIST and 11.60-point gap on CIFAR-10 at $\alpha = 1.0$, reflecting the soft-label threat model scope. The defense targets soft-label knowledge distillation, where margin-adaptive perturbations directly reduce surrogate fidelity. The fundamental trade-off—preserving argmax to guarantee zero accuracy loss—means CPLS provides *no* protection against hard-label extraction attacks that only observe top-1 predictions. Hard-label threats are addressed separately through query detection (e.g., PRADA) or rate limiting, which complement CPLS in a layered architecture.

Adaptive Attacks. CPLS resists naive EOT averaging due to deterministic perturbations. The orthogonal projection property makes direct gradient-based inversion mathematically challenging: recovering the original logits z from perturbed logits z' requires solving an underdetermined system (inferring K unknowns from $K - 1$

independent equations, since $\delta_y = 0$). Adversarial data augmentation to corrupt the input-keyed hash is similarly constrained, as the deterministic mapping ensures consistent perturbations regardless of query repetition. Attackers with knowledge of the hash function and seed face a reduced barrier to reconstructing perturbations for known inputs. **Hash seed rotation**, periodically changing the server-side hash seed based on time intervals or query counts, limits the temporal window during which reconstructed perturbations remain valid, analogous to API key rotation in access control systems. Cryptographic hardening (e.g., HMAC-based hashing with server-secret keys) provides complementary protection by making seed extraction computationally infeasible.

Data-Free Extraction. Data-free extraction attacks that synthesize training data via generative models [16] represent a complementary threat model. CPLS targets probability-returning APIs under soft-label knowledge distillation; data-free approaches operate in a fundamentally different threat space. The deterministic hash embedding resists EOT averaging within the soft-label setting. Combining CPLS with input validation (e.g., detecting out-of-distribution queries) or query monitoring provides layered defense across both query-based and data-free extraction threat models.

Scalability. The per-sample margin computation and hash embedding add minimal overhead ($< 1\%$ latency increase), making CPLS practical for deployment in production APIs.

Wireless Deployment. In wireless ML-as-a-service scenarios, CPLS's sub-millisecond overhead and zero-accuracy-loss guarantee are particularly valuable. Edge-deployed spectrum sensing models cannot tolerate accuracy degradation due to regulatory requirements (e.g., FCC Part 15 compliance for unlicensed spectrum use), and CPLS's deterministic nature avoids the stochastic overhead that complicates real-time wireless processing pipelines. The method's input-keyed hashing is directly applicable to structured wireless inputs (e.g., IQ samples from software-defined radios, spectral power features) without modification. Our spectrum sensing evaluation (Table 6) demonstrates this generality to non-image domains, confirming suitability for diverse wireless security applications including RF fingerprinting and intrusion detection.

6 Conclusion

CPLS provides a theoretically grounded, lightweight mechanism to reduce soft-label model stealing effectiveness without sacrificing clean accuracy. Our evaluations across MNIST, CIFAR-10, and spectrum sensing demonstrate up to 13.1% relative mutual information reduction with zero accuracy loss and perfect consistency preservation, alongside strong resistance to EOT averaging attacks. The deterministic, margin-adaptive design offers analytical guarantees while providing practical extraction resistance suitable for deployment in safety-critical wireless ML-as-a-service applications (e.g., spectrum sensing, RF fingerprinting) where prediction consistency is non-negotiable.

We position CPLS as a first line of defense in a layered security architecture: it degrades the soft-label extraction channel while preserving service utility, complementing query-rate defenses (e.g., PRADA) that address hard-label threats and watermarking techniques that enable post-hoc ownership verification. The trade-off—abandoning hard-label protection to guarantee zero accuracy

loss—is explicit and justified for domains where misclassification carries regulatory or safety consequences.

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